

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12418283
Kind Code	B2
Date of Patent	September 16, 2025
Inventor(s)	Saputra; Nitz et al.

Programmable self-biasing inverter for correcting duty cycle distortion

Abstract

A clock conditioning circuit includes a duty cycle correction circuit. The duty cycle correction circuit has a first input capacitor, a first self-biasing inverter and a variable capacitor. The first self-biasing inverter has an input coupled to the first input capacitor. The variable capacitor may be coupled to the first input capacitor. The variable capacitor may be configured to receive a first clock signal. A capacitance of the variable capacitor may be programmable by a capacitance control signal.

Inventors:	Saputra; Nitz (San Diego, CA), Wadhwa; Sameer (San Diego, CA)
Applicant:	QUALCOMM Incorporated (San Diego, CA)
Family ID:	1000008820260
Assignee:	QUALCOMM INCORPORATED (San Diego, CA)
Appl. No.:	18/391096
Filed:	December 20, 2023

Prior Publication Data

Document Identifier	Publication Date
US 20250211218 A1	Jun. 26, 2025

Publication Classification

Int. Cl.:	H03K5/156 (20060101)
U.S. Cl.:	
CPC	H03K5/1565 (20130101);

Field of Classification Search

USPC: None

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
8653856	12/2013	Piepenstock	327/108	H03K 19/018528
2012/0081179	12/2011	Visser	330/251	H03F 1/02
2022/0209716	12/2021	Yamamoto	N/A	H03B 5/368

Primary Examiner: Cox; Cassandra F

Attorney, Agent or Firm: Loza & Loza, LLP

Background/Summary

TECHNICAL FIELD

(1) The present disclosure generally relates to a clock generation circuit and, more particularly, to circuitry for conditioning clock signals used by a serializer.

BACKGROUND

(2) Electronic device technologies have seen explosive growth over the past several years. For example, growth of cellular and wireless communication technologies has been fueled by better communications, hardware, larger networks, and more reliable protocols. Wireless service providers are now able to offer their customers an ever-expanding array of features and services, and provide users with unprecedented levels of access to information, resources, and communications. To keep pace with these service enhancements, mobile electronic devices (e.g., cellular phones, tablets, laptops, etc.) have become more powerful and complex than ever. Wireless devices may include a high-speed bus interface for communication of signals between hardware components.

(3) High-speed serial buses offer advantages over parallel communication links when, for example, there is demand for reduced power consumption and smaller footprints in integrated circuit (IC) devices. In a serial interface, data is converted from parallel words to a serial stream of bits using a serializer and is converted back to parallel words at the receiver using a deserializer. For example, the high-speed bus interface may be implemented using a Peripheral Component Interconnect Express (PCIe) bus, Universal Serial Bus (USB) or Serial Advanced Technology Attachment (SATA), among others.

(4) IC devices may include a serializer/deserializer (SERDES) to transmit and receive through a communication link. In high-speed applications, timing of the operation of a SERDES may be controlled by multiple clock signals. The use of multiple increased frequency clock signals results in increased power consumption. Furthermore, the SERDES generally operates using clock signals that have the same frequency but different phases. Performance, accuracy or reliability of the SERDES may depend on frequency and the phase relationships of the clock signals and drift or other variations in the phase relationships of clock signals can result in errors in received or transmitted data. Therefore, there is an ongoing need improve, low-power clock generation and clock conditioning circuits for use in interfaces that employ SERDES circuits.

SUMMARY

- (5) Certain aspects of the disclosure relate to IC devices that include a bus interface. The bus interface may include circuits that can be configured to condition clock signals to ensure accurate and reliable control of transmissions over a communication bus or for accurate and reliable capture of data from signals transmitted over the communication bus.
- (6) In various aspects of the disclosure, a clock conditioning circuit includes a duty cycle correction circuit, that includes a first input capacitor and a first self-biasing inverter that has an input coupled to the first input capacitor. The clock conditioning circuit includes a variable capacitor coupled to the first input capacitor. The variable capacitor is configured to receive a first clock signal. A capacitance of the variable capacitor is programmable by a capacitance control signal.
- (7) In various aspects of the disclosure, a method for conditioning clock signals includes providing a first clock signal to an input of a duty cycle correction circuit, the duty cycle correction circuit having a first self-biasing inverter and a first input capacitor, and selecting a capacitance value provided by a variable capacitor that is coupled to the first input capacitor. A capacitance provided by the variable capacitor may be programmable by a capacitance control signal.
- (8) In various aspects of the disclosure, an apparatus includes means for correcting a duty cycle of a clock signal, including a first self-biasing inverter and a first input capacitor. The apparatus includes means for adjusting transition times of edges in the clock signal, including a variable capacitor that is coupled to the first input capacitor. The apparatus includes means for programming a capacitance value provided by a variable capacitor. The means for programming the capacitance value may be responsive to a capacitance control signal.
- (9) In certain aspects, the clock conditioning circuit includes a second input capacitor, and a second self-biasing inverter. The second self-biasing inverter has an input coupled to the second clock signal through the second input capacitor. The variable capacitor is coupled between the first clock signal and the second clock signal. The clock conditioning circuit may include an input circuit configured to receive a single-ended input clock signal and to convert the single-ended input clock signal to a differential clock signal. The differential clock signal includes the first clock signal and the second clock signal. A skew correction circuit may be provided between the first clock signal and the second clock signal and may include at least one pair of back-to-back inverters. The variable capacitor may include a plurality of switches controlled by the capacitance control signal. Each switch may be configured to couple the first clock signal and the second clock signal through a corresponding pair of series connected differential capacitors when activated.
- (10) In one aspect, the capacitance control signal is a multi-bit unary or binary encoded signal. In one aspect, the first self-biasing inverter includes a feedback resistor coupled between the input of the first self-biasing inverter and an output of the first self-biasing inverter.
- (11) In certain aspects, the clock conditioning circuit includes a configurable inverter configured to provide the first clock signal by delaying an input clock signal. In this configuration, the variable capacitor may be coupled between the first clock signal and ground. The variable capacitor may include a plurality of switches controlled by the capacitance control signal. Each switch may be configured to couple the first clock signal to ground through one of a plurality of capacitors provided in the variable capacitor when the each switch is activated.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 illustrates an example of a system-on-a-chip in accordance with certain aspects of the present disclosure.
- (2) FIG. 2 illustrates an apparatus employing interconnected chiplets.
- (3) FIG. 3 illustrates an example of a system that employs a multi-channel data communication

link.

(4) FIG. 4 illustrates an example of a conventional quarter-rate transmitter.

(5) FIG. 5 illustrates examples of clock conditioning circuits.

(6) FIG. 6 illustrates certain aspects of a programmable clock conditioning circuit that has been configured in accordance with certain aspects of this disclosure.

(7) FIG. 7 illustrates certain aspects of a single-ended programmable clock conditioning circuit that has been configured in accordance with certain aspects of this disclosure.

(8) FIG. 8 is a flowchart of a method for generating clock signals in accordance with certain aspects of this disclosure.

DETAILED DESCRIPTION

(9) The detailed description set forth below in connection with the appended drawings is intended as a description of various configurations and is not intended to represent the only configurations in which the concepts described herein may be practiced. The detailed description includes specific details for the purpose of providing a thorough understanding of various concepts. However, it will be apparent to those skilled in the art that these concepts may be practiced without these specific details. In some instances, well-known structures and components are shown in block diagram form in order to avoid obscuring such concepts.

(10) Several aspects of the invention will now be presented with reference to various apparatus and methods. These apparatus and methods will be described in the following detailed description and illustrated in the accompanying drawings by various blocks, modules, components, circuits, steps, processes, algorithms, etc. (collectively referred to as “elements”). These elements may be implemented using electronic hardware, computer software, or any combination thereof. Whether such elements are implemented as hardware or software depends upon the particular application and design constraints imposed on the overall system.

(11) Data communication links employed by SoCs and other IC devices to connect processors with modems and other peripherals may be operated in accordance with industry or proprietary standards or protocols associated with certain functions or types of devices. According to certain aspects of the disclosure, a serial data link may be used to interconnect electronic devices that are subcomponents of an apparatus such as a cellular phone, a smart phone, a session initiation protocol (SIP) phone, a laptop, a notebook, a netbook, a smartbook, a personal digital assistant (PDA), a satellite radio, a global positioning system (GPS) device, a smart home device, intelligent lighting, a multimedia device, a video device, a digital audio player (e.g., MP3 player), a camera, a game console, an entertainment device, a vehicle component, a wearable computing device (e.g., a smart watch, a health or fitness tracker, eyewear, etc.), an appliance, a sensor, a security device, a vending machine, a smart meter, or any other similar functioning device.

(12) Certain aspects of the disclosure are applicable to a wide variety of input/out (I/O) circuits. In one example, certain aspects of this disclosure may be implemented in I/O circuits that provide an interface between core circuits and memory devices. Many mobile devices employ Synchronous Dynamic Random Access Memory (SDRAM), including Low-Power Double Data Rate SDRAM, which may be referred to as low-power DDR SDRAM, LPDDR SDRAM or, in some instances, LPDDR_x where x describes the technology generation of the LPDDR SDRAM. Later generations of LPDDR SDRAM designed to operate at higher operating frequencies may employ lower voltage levels in the core of an SoC or memory device to mitigate for increased power associated with the higher operating frequencies.

(13) Process technology employed to manufacture semiconductor devices, including IC devices is continually improving. Process technology includes the manufacturing methods used to make IC devices and defines transistor size, operating voltages and switching speeds. Features that are constituent elements of circuits in an IC device may be referred as technology nodes and/or process nodes. The terms technology node, process node, process technology may be used to characterize a specific semiconductor manufacturing process and corresponding design rules. Faster and more

power-efficient technology nodes are being continuously developed through the use of smaller feature size to produce smaller transistors that enable the manufacture of higher-density ICs.

(14) Certain examples of circuits used to generate, process or condition clock signals are disclosed herein. Certain of these circuits are illustrated as being implemented using certain combinations of P-type metal-oxide-semiconductor (PMOS) transistors and N-type metal-oxide-semiconductor (NMOS) transistors. These circuits are provided by way of example only, and it is contemplated that the concepts disclosed herein can be implemented in circuits that use different combinations of NMOS and PMOS transistors. Circuits that include NMOS and PMOS transistors are typically coupled to the rails of a power supply. The power supply provides a current that flows from a higher voltage rail to a lower voltage rail. A rail may include some combination of conductors, wires, connectors and other types of interconnect. For the purposes of this description, the higher voltage rail may be referenced as “VDD” or “V.sub.DD” and the lower voltage rail may be referred to as Ground. In some implementations, power may be provided to certain circuits through more than two rails.

(15) For the purposes of this disclosure, a half-rate clock signal used to control operation of SERDES circuits has a frequency that is half the frequency of the data signal. The term quarter-rate as used herein refers to the use of a multiphase clock signal with a frequency that is a quarter of the frequency of the data signal. In-phase signal and quadrature versions of a quarter-rate clock signal may be used in certain SERDES circuits. The quadrature version of the clock signal is phase-shifted with respect to the in-phase version of the clock signal by 90°.

(16) FIG. 1 illustrates an example of an apparatus **100** in which certain components and interconnections are implemented in an SoC. The SoC may include or be coupled to a memory interface/bus **126** that can be adapted according to certain aspects of the present disclosure. The apparatus **100** may include a number of heterogeneous processors, such as a central processing unit (CPU) **102**, a modem processor **104**, a graphics processor **106**, and an application processor **108**. Each processor **102**, **104**, **106**, **108**, may include one or more cores, and each processor/core may perform operations independent of the other processors/cores. The processors **102**, **104**, **106**, **108** may be organized in close proximity to one another (e.g., on a single substrate, die, integrated chip, etc.) so that the processors may operate at a much higher frequency/clock rate than would be possible if the signals were to travel off-chip. The proximity of the cores may also allow for the sharing of on-chip memory and resources (e.g., voltage rails), as well as for more coordinated cooperation between cores.

(17) The apparatus **100** may include system components and resources **110** for managing sensor data, analog-to-digital conversions, and/or wireless data transmissions, and for performing other specialized operations (e.g., decoding high-definition video, video processing, etc.). System components and resources **110** may also include components such as voltage regulators, oscillators, phase-locked loops (PLLs), peripheral bridges, data controllers, system controllers, access ports, timers, and/or other similar components used to support the processors and software clients running on the computing device. The system components and resources **110** may also include circuitry for interfacing with peripheral devices, such as cameras, electronic displays, wireless communication devices, external memory chips, etc.

(18) The apparatus **100** may further include a serial bus controller **112** such as a Universal Serial Bus (USB) controller, one or more memory controllers **114**, and a centralized resource manager (CRM) **116**. The apparatus **100** may also include an input/output module (not illustrated) for communicating with resources external to the SoC, each of which may be shared by two or more of the internal SoC components.

(19) The processors **102**, **104**, **106**, **108** may be interconnected to the serial bus controller **112**, the memory controller **114**, system components and resources **110**, CRM **116**, and/or other system components via an interconnection/bus module **122**, which may include an array of reconfigurable logic gates and/or implement a bus architecture. Communications may also be provided by

advanced interconnects, such as high-performance networks on chip (NoCs).

(20) The interconnection/bus module **122** may include or provide a bus mastering system configured to grant SoC components (e.g., processors, peripherals, etc.) exclusive control of the bus (e.g., to transfer data in burst mode, block transfer mode, etc.) for a set duration, number of operations, number of bytes, etc. In some cases, the interconnection/bus module **122** may implement an arbitration scheme to prevent multiple master components from attempting to drive the bus simultaneously. The memory controller **114** may be a specialized hardware module configured to manage the flow of data to and from a memory **124** via the memory interface/bus **126**.

(21) The memory controller **114** may comprise one or more processors configured to perform read and write operations with the memory **124**. Examples of processors include microprocessors, microcontrollers, digital signal processors (DSPs), field programmable gate arrays (FPGAs), programmable logic devices (PLDs), state machines, gated logic, discrete hardware circuits, and other suitable hardware configured to perform the various functionality described throughout this disclosure. In certain aspects, the memory **124** may be part of the apparatus **100**.

(22) FIG. 2 illustrates an example of an apparatus **200** in which certain components are implemented using multiple chiplets that are interconnected using one or more data communication buses. In one example, the apparatus **200** may be enclosed within a wearable device a portable or wearable processing and/or communication device (each of which being referred to herein as a portable communication device or PCD), sensors, instruments, appliances and other such devices include one or more ICs. These devices may include mobile phones, tablet computers, palmtop computers, portable digital assistants (PDAs), portable game consoles, and other portable electronic devices such as the illustrated smartwatch **210**. PCDs commonly contain integrated circuits or SoCs that include numerous components or subsystems designed to work together to deliver functionality to a user. The various SoC subsystems may communicate with each other via one or more intra-chip data buses or similar data communication interconnects. PCDs may have multiple SoCs that communicate with each other via similar inter-chip interconnects. The ICs are typically packaged in an IC package, which may be referred to as a “semiconductor package” or “chip package.” The IC package typically includes a package substrate and one or more IC chips or other electronic modules mounted to the package substrate to provide electrical connectivity to the IC chips. For example, an IC chip in an IC package may be configured as an SoC. The IC chips are electrically coupled to other IC chips and/or to other components in the IC package through electrical coupling to metal lines in the package substrate. The IC chips can also be electrically coupled to other circuits outside the IC package through electrical connections of external metal interconnects (e.g., solder bumps) of the IC package.

(23) Chiplet technology can be used to address some of the performance, power and size design requirements for complex SoCs used in certain mobile or wearable devices. The block diagram in FIG. 2 illustrates certain aspects of an apparatus **200** that can be constructed using chiplets. The apparatus **200** may be configured by selecting a combination of chiplets that implement certain subsystems or distinct functional elements. In the illustrated example, the apparatus **200** includes a set of primary chiplets **202** that enable the apparatus **200** to perform core processing, security and communication functions. The set of primary chiplets **202** include a processor, memory and one or more modems. The illustrated apparatus **200** also includes a set of application-specific chiplets **204** that includes an application processor, display driver, camera interface and audio controller. In a remote sensing device or appliance, the audio-visual components could be omitted and may be replaced with analog-to-digital controllers, for example.

(24) The apparatus **200** may include a variety of processing engines, such as central processing units (CPUs) with multiple cores, graphical processing units (GPUs), digital signal processors (DSPs), neural processing units (NPU), wireless transceiver units (also referred to as modems), peripherals, display and imaging interfaces, etc. Each of these subsystems and other functional

elements can be implemented as an individual chiplet, or as a combination of chiplets. The chiplets included in the apparatus **200** can be proprietary or may be acquired from a variety of sources. An SoC may be constructed from chiplets manufactured at different process nodes and/or operated at different voltages.

(25) FIG. **3** illustrates an example of a system that employs a multi-channel data communication link **380** to couple a modem **300** with a wireless transceiver **340**. The data communication link **380** includes data channels **382** and **386** and a clock channel **384** that provide a transmission medium through which signals propagate between devices. In the illustrated example, a modem **300** transmits data in a first signal over a first data channel **382** to a wireless transceiver **340** and receives data in a second signal transmitted over a second data channel **386**. Data signals are transmitted over the data channels **382** and **386** in accordance with timing information provided by a bus clock signal **330** transmitted over the clock channel **384**.

(26) The modem **300** may include a serializer **302** configured to convert n-bit parallel data elements, bytes or words into a serial data stream for transmission in a transmit data signal **322** over the first data channel **382**. The transmit data signal **322** may be preconditioned by a pre-equalizing circuit, such as the illustrated digital feed-forward equalizer (the FFE **304**), in order to combat or compensate for signal distortions attributable to inter-symbol interference (ISI), reflection and other effects that can be expected to limit bandwidth in first data channel **382**. The preconditioned transmit data signal **324** output by the FFE **304** is provided to a driver circuit **306** that is configured drive the first data channel **382**.

(27) The modem **300** may include a serializer **302** configured to convert n-bit parallel data elements, bytes or words into a serial data stream for transmission in a serialized data signal **322**. The serialized data signal **322** may be preconditioned by a pre-equalizing circuit, such as the illustrated digital feed-forward equalizer (the FFE **304**), in order to combat or compensate for signal distortions attributable to inter-symbol interference (ISI), reflection and other effects that can be expected to limit bandwidth in the first data channel **382**. A preconditioned data signal **324** output by the FFE **304** is provided to a driver circuit **306** that is configured generate and transmit a differential transmit data signal **326** over the first data channel **382**. For the purposes of this disclosure, a differential signal includes two complementary signals that are phase-shifted by 180° with respect to one another.

(28) The wireless transceiver **340** can be configured to process a data signal **360** received over the first data channel **382**. The data signal **360** may be provided to a differential receiver **342**, which may include or cooperate with an equalizing circuit. In one example, continuous time linear equalization (CTLE) may be used to compensate for certain losses experienced in the first data channel **382**. The first data channel **382** may be characterized in some respects as a low-pass filter. In the illustrated example, the differential receiver **342** outputs an equalized data signal **362** that is sampled by a slicer **344**. The slicer **344** may be implemented using a D-flipflop or the like and may be configured to capture signaling state of the equalized data signal **362** under the control of edges in a sampling clock signal **372** generated by a clock and data recovery (CDR) circuit **348**. The output of the slicer **344** may be provided to a deserializer **346** that is clocked in accordance with one or more clock signals provided by the CDR circuit **348**. The CDR circuit **348** may be configured to delay or phase shift a receiver clock signal **370** to ensure that edges in the sampling clock signal **372** are timed to optimize sampling reliability.

(29) In the illustrated wireless transceiver **340**, the receiver clock signal **370** is derived from a received bus clock signal **374** from the clock channel **384**. A differential receiver **352** coupled to the clock channel **384** may be configured to equalize the received bus clock signal **374**, and a duty cycle correction circuit **350** may be used to adjust the duty cycle of the receiver clock signal **370**. The receiver clock signal **370** is provided to a serializer **354** that is configured to convert n-bit parallel data elements, bytes or words into a serial data stream for transmission in a serialized data signal **364**. The serialized data signal **364** may be preconditioned by a pre-equalizing circuit, such

as the illustrated FFE **356**, in order to combat or compensate for signal distortions attributable to ISI, reflection and other effects that can be expected to limit bandwidth in the second data channel **386**. A preconditioned data signal **366** output by the FFE **356** is provided to a driver circuit **358** that is configured generate and transmit a differential transmit data signal **368** over the second data channel **386**.

(30) The illustrated modem **300** can be configured to process a data signal **332** received over the second data channel **386**. The data signal **332** may be provided to a differential receiver **320**, which may include or cooperate with an equalizing circuit. In one example, CTLE may be used to compensate for certain losses experienced in the second data channel **386**. The second data channel **386** may be characterized in some respects as a low-pass filter. In the illustrated example, the differential receiver **320** outputs an equalized data signal **328** that is sampled by a slicer **318**. The slicer **318** may be implemented using a D-flipflop or the like and may be configured to capture signaling state of the equalized data signal **328** under the control of edges in a sampling clock signal **334** generated by a CDR circuit **314**. The output of the slicer **318** may be provided to a deserializer **316** that is clocked in accordance with one or more clock signals provided by the CDR circuit **314**. The CDR circuit **314** may be configured to delay or phase shift a transmitter clock signal to ensure that edges in the sampling clock signal **334** are timed to optimize sampling reliability.

(31) A clock generation circuit, including the illustrated phase locked loop **308**, may generate multiple clock signals **336a**, **336b**, **336c** used by the modem **300**. One or more of the clock signals **336a**, **336b**, **336c** may be a divided version of a base clock signal generated by the PLL **308**. One or more of the clock signals **336a**, **336b**, **336c** may be phase shifted with respect to the base clock signal. In one example, the serializer **302** may produce the serialized data signal **322** using timing provided by a first clock signal **336a**. In another example, the bus clock signal **330** transmitted over the clock channel **384** may be derived from a second clock signal **336b**. In some instances, a duty cycle correction circuit **310** may be used to adjust the duty cycle of the second clock signal **336b** and to provide an input to a driver circuit **312** that is configured drive the clock channel **384**. In another example, the CDR circuit **348** may generate the sampling clock signal **334** from a third clock signal **336c**.

(32) Limiting power consumption presents a major challenge in communication interfaces, including communication interfaces that include SERDES circuits. In mobile communication devices, reducing power consumption can increase battery life between charges. Accordingly, power consumption is a parameter that must be considered when physical layer (PHY) circuits are designed for communication interfaces that are required to meet ever-increasing demands for data rates and corresponding signaling rates associated with the communication interface.

(33) Demands for higher data rates and increased performance from SERDES-based PHY circuits are, in part, a consequence of continual advances in process technology and changing industry and proprietary standards. Increased switching frequency of PHY circuits in communication interfaces require accurate clock signals that are used to serialize or deserialize a data stream. Clock signals transmitted over a communication link are subject to distortion, delays and skews that require phase correction, alignment and other adjustments to timing. As used herein, a clock conditioning circuit may be used for phase correction, signal equalization, signal alignment, reduction or removal of skew between complementary signals, duty cycle correction, other adjustments to frequency and timing of a clock signal. A device, such as the wireless transceiver **340** illustrated in FIG. **3**, typically includes conditioning circuits to correct a variety of timing distortions associated with a clock signal that is received over a communication link to ensure accuracy of data capture from received data signals and to ensure the timing accuracy of data signals to be transmitted over the communication link.

(34) In a typical serializer, for example, the clock input duty-cycle and skew between the complementary signals in a differential clock signal are critical to the timing accuracy of the

serializer output. For the purposes of this disclosure, a differential signal includes two complementary signals that are phase-shifted by 180° with respect to one another. It is generally difficult for a designer to anticipate or quantify the large number of parameters that can affect the quality of a received clock signal and designers may be required to design complex receiver circuits and clock conditioning circuits, resulting in higher die area and power consumption. Examples of these clock conditioning circuits include duty cycle correction circuits, skew correction circuits and phase shifting circuits.

(35) FIG. 4 illustrates an example of a quarter-rate transmitter **400**. A quarter-rate transmitter may be sensitive to duty cycle error and clock skew. The quarter-rate transmitter **400** includes a serializer **402** that responds to edges provided in a multiphase clock signal **410** to provide a data output signal **440** that includes a serial bitstream. The serial bitstream is obtained by serializing a multibit data input **420**. In the illustrated example, the multibit data input **420** is provided to a first serializing circuit **412** that receives a quantity (n) of bits and provides a 4-bit datastream **422** that includes a sequence of 4-bit units of data. In some implementations, the 4-bit datastream **422** may be generated using one or more multiplexers, one or more shift registers, or some combination of multiplexers and shift registers. In some implementations, the 4-bit datastream **422** may be received from registers written by a processor. The 4-bit datastream **422** is provided to a second serializing circuit **414** that provides the data output signal **440** according to timing provided by edges in the multiphase clock signal **410**.

(36) The multiphase clock signal **410** is derived from a quarter-rate input clock signal **430**. In the illustrated example, the quarter-rate input clock signal **430** is provided to a duty cycle correction (DCC) circuit **404** that can be configured to ensure that the duration between rising edges and subsequent falling edges in a corrected clock signal **432** is substantially the same (i.e., within predefined tolerances) as the duration between falling edges and subsequent rising edges in the corrected clock signal **432**. In some implementations, a divider circuit that provides the quarter-rate input clock signal **430** may be configured to ensure that the duty cycle of the quarter-rate input clock signal **430** lies within tolerances defined for the quarter-rate transmitter **400**.

(37) The corrected clock signal **432** may be provided to a buffer circuit **418** that outputs a differential in-phase clock signal **438**. The corrected clock signal **432** is further provided to a delay circuit **406** that outputs a delayed clock signal **434**. The delay circuit **406** can be configured by a control signal **442** to delay the corrected clock signal **432** by a duration that corresponds to 25% of the period of the quarter-rate input clock signal **430** in order to cause the delayed clock signal **434** to be phase-shifted by 90° with respect to the corrected input clock signal **432**. The delayed clock signal **434** may be provided to a buffer circuit **416** that outputs a differential quadrature clock signal **436**. In one example, the delay duration provided by the delay circuit **406** may be selected or adjusted using a coded value provided by the control signal **442**. In another example, the control signal **442** may provide feedback that adjusts the delay duration provided by the delay circuit **406**.

(38) The differential in-phase clock signal **438** (I) and the differential quadrature clock signal **436** (Q) may be provided to conditioning circuits **408**. The conditioning circuits **408** may include clock distribution drivers, buffers, additional DCC circuits and the like. The conditioning circuits **408** output the multiphase clock signal **410**.

(39) FIG. 4 includes a timing diagram **450** that illustrates duty cycle distortion in a clock signal (the Clk.sub.DCD signal **456**) and skew between complementary signals (the Clk signal **452** and Clk_B signal **454**) in a differential clock signal. For the purposes of this disclosure, duty cycle may be defined as the percentage or fraction of a clock cycle during which a signal is in an active state, or has a voltage greater than the mid-level of the voltage range in which the signal varies. The complementary signals in a differential signal can have different duty cycles. Duty cycle distortion can affect a single-ended clock or differential clock signals. In the illustrated example, the duty cycle of the digital Clk.sub.DCD signal **456** results in a difference **458** in duration between the active and inactive signaling states.

(40) For the purposes of this disclosure, the terms “skew” and “clock skew” may be defined as the difference in time of arrival of related signals at respective inputs of a circuit. Skew may be caused by differences in propagation delays experienced by the related signals. Propagation delays may be caused by differences in physical path lengths or in communication channel impedances, by the deployment of different numbers of intervening circuits between the source and destination of the related signals, by jitter caused by noise or electromagnetic interference, and/or by differences introduced by variations in manufacturing processes, voltage or temperature (PVT). In the illustrated example, skew **460** is shown between the Clk signal **452** and Clk_B signal **454**.

(41) FIG. 5 illustrates clock conditioning circuits **500**, **520** that may be implemented in some systems. The clock conditioning circuit **500** represents a conservative design for duty cycle correction and includes a sufficient number of duty cycle correction stages to accommodate expected duty cycle distortion given a limited correction capability provided by each duty cycle correction stage. The number and configuration of the duty cycle correction stages may be selected to correct the expected worst case duty cycle distortion in an input clock signal and to compensate for duty cycle distortion introduced within the clock conditioning circuit **500**.

(42) The first clock conditioning circuit **500** receives a single-ended clock signal (Clk_SE **508**) and outputs a differential clock signal that includes a pair of complementary output signals **510a**, **510b**. In the illustrated example, Clk_SE **508** is coupled to a single-ended to differential signal conversion circuit **504** through an input duty cycle correction circuit **502**. The input duty cycle correction circuit **502** provides two stages of duty cycle correction. Each stage of the input duty cycle correction circuit **502** includes a capacitor **518.sub.1**, **518.sub.2** and a self-biasing inverter **512.sub.1**, **512.sub.2**. Complementary clock skew that may be generated in the single-ended to differential signal conversion circuit **504** can be corrected using a skew correction circuit **516** that includes a pair of back-to-back inverters coupled in parallel between the complementary signals **514a**, **514b** output by the single-ended to differential signal conversion circuit **504**. The input of each back-to-back inverter is coupled to the output of the other back-to-back inverter. A two-stage output duty cycle correction circuit **506** receives the complementary signals **514a**, **514b** output by the single-ended to differential signal conversion circuit **504** and drives the complementary output signals **510a**, **510b**. The duty cycle of a first complementary output signal **510a** is corrected by a two-stage subcircuit in which each stage includes a capacitor **518.sub.3**, **518.sub.4** and a self-biasing inverter **512.sub.3**, **512.sub.4**. The duty cycle of a second complementary output signal **510b** is corrected by a two-stage subcircuit in which each stage includes a capacitor **518.sub.5**, **518.sub.6** and a self-biasing inverter **512.sub.5**, **512.sub.6**.

(43) The second clock conditioning circuit **520** represents a conservative design for clock skew correction. The second clock conditioning circuit **520** includes a sufficient number of skew correction stages to correct the expected worst-case skew between the complementary signals **514a**, **514b**. The number and configuration of the skew correction stages may be selected to correct the clock skew and reduce duty cycle distortion in an input clock signal.

(44) The second clock conditioning circuit **520** receives a single-ended clock signal (Clk_SE **528**) and outputs a differential clock signal that includes a pair of complementary output signals **540a**, **540b**. In the illustrated example, Clk_SE **528** is coupled to a single-ended to differential signal conversion circuit **522**, which has a differential output that is coupled to three skew correction circuits **526a**, **526b**, **526c**. In some implementations, a greater number of skew correction stages may be required. Each of the skew correction circuits **526a**, **526b**, **526c** includes a pair of back-to-back inverters coupled in parallel between a corresponding pair complementary signals. In the illustrated example, the back-to-back inverters of a first skew correction circuit **526a** is coupled between the complementary signals **536a**, **536b** output by the single-ended to differential signal conversion circuit **522**, a second skew correction circuit **526b** is coupled between the complementary signals **538a**, **538b** and a third skew correction circuit **526c** is coupled between the complementary output signals **540a**, **540b**.

(45) The illustrated single-ended to differential signal conversion circuit **522** provides two signal paths for Clk_SE **528**. The first signal path is a non-inverting path that includes two series-connected inverters **530a**, **530b**. The second signal path is an inverting path that includes a single inverter **534** coupled in series with a pass gate circuit **532**. The pass gate circuit **532** may be configured to help match delays in the two signal paths through the single-ended to differential signal conversion circuit **522**.

(46) The number of subcircuits and stages needed to implement the clock conditioning circuits **500**, **520** illustrated in FIG. 5 requires the occupation of significant die real estate and can increase power consumption due to the number of circuits that switch at clock frequency.

(47) Certain aspects of this disclosure relate to programmable clock conditioning circuits that can be used to improve clock duty cycle distortion correction without the need for calibration. Clock conditioning circuits provided in accordance with certain aspects of this disclosure can be implemented in transmitter and receivers used in communication links. Clock conditioning circuits provided in accordance with certain aspects of this disclosure can be used to reduce or eliminate duty cycle distortion and skew in clock signals that are used to control serializing or deserializing circuits in a communication interface.

(48) FIG. 6 illustrates certain aspects of a programmable clock conditioning circuit **600** that has been configured in accordance with certain aspects of this disclosure. The programmable clock conditioning circuit **600** includes a skew correction circuit **604**, a variable capacitor **606** and a duty cycle correction circuit **608**. The programmable clock conditioning circuit **600** receives a single-ended input clock signal (Clk_SE signal **610**) and outputs a differential clock signal that includes a pair of complementary output signals **620a**, **620b**. In the illustrated example, the Clk_SE signal **610** is coupled to a single-ended to differential signal conversion circuit **602**. The skew correction circuit **604** includes a pair of back-to-back inverters coupled in parallel between the complementary signals **614a**, **614b** output by the single-ended to differential signal conversion circuit **602**. The input of each back-to-back inverter is coupled to the output of the other back-to-back inverter. The duty cycle correction circuit **608** receives the complementary signals **620a**, **620b** and has a differential output that is coupled to a pair of inverters **612a**, **612b**. The inverters **612a**, **612b** provide the complementary output signals **620a**, **620b**.

(49) The duty cycle correction circuit **608** is programmable. A first complementary signal **614a** received from the single-ended to differential signal conversion circuit **602** is coupled through a first capacitor **618a** to a first self-biasing inverter **628a**. The second complementary signal **614b** received from the single-ended to differential signal conversion circuit **602** is coupled through a second capacitor **618b** to a second self-biasing inverter **628b**. Capacitors **618a**, **618b** block the direct current (DC) component of their respective complementary signals **614a**, **614b** such that the signals received at the inputs **616a**, **616b** to the self-biasing inverters **628a**, **628b** are centered on a voltage level that is nominally the mid-point voltage level of the corresponding complementary signals **614a**, **614b**. A resistance provided in each of the self-biasing inverters **628a**, **628b** defines a resistance-capacitance (RC) constant that determines the rise time of edges in the signals passed through the capacitors **618a**, **618b**.

(50) FIG. 6 includes an example of a self-biasing inverter **630** that may be used to implement the self-biasing inverters **628a**, **628b** in the programmable clock conditioning circuit **600**. The self-biasing inverter **630** receives a single-ended input clock signal (the Inv.sub.inp signal **640**) and provides a single-ended output clock signal (the Inv.sub.out signal **644**). The self-biasing inverter **630** includes a pair of transistors **636a**, **636b** that are configured to operate as a push-pull amplifier that drives the Inv.sub.out signal **644**. The Inv.sub.inp signal **640** is coupled to the gate of each of the pair of transistors **636a**, **636b** through an input capacitor **632**. The DC component of the Inv.sub.inp signal **640** is blocked by the input capacitor **632** to provide a clock signal (the Inv.sub.mid signal **642**) that is centered on the mid-level DC voltage of the Inv.sub.inp signal **640**. A feedback resistor **634** is coupled between the Inv.sub.out signal **644** and the gates of the pair of

transistors **636a**, **636b**. The feedback resistor **634** is configured to automatically bias the pair of transistors **636a**, **636b**.

(51) The RC constant provided by the feedback resistor **634** and the input capacitor **632** is selected to provide a maximum transition time for signals in the Inv.sub.mid signal **642**. The timing diagram **650** provides an example of the timing of signals associated with the self-biasing inverter **630**. The Inv.sub.inp signal **640** is depicted as having relatively fast transitions between signaling states **646a**, **646b**. In the illustrated example, duty cycle is not at 50% as would be desired since the low signaling state **646a** has a longer duration than the high signaling state **646b**. The mid-level voltage of the Inv.sub.mid signal **642** rises above the mid-level voltage of the Inv.sub.inp signal **640** due to the imbalance of charge accumulation during the signaling states **646a**, **646b**.

(52) The rise in the Inv.sub.mid signal **642** tends to drive the duty cycle of the Inv.sub.out signal **644** towards 50%, reflecting a more balanced charge accumulation between signaling states **646a**, **646b**. The raised voltage levels of the Inv.sub.mid signal **642** with respect to the Inv.sub.inp signal **640** changes the relationship of the Inv.sub.mid signal **642** with the switching thresholds used by the inverters **628a**, **628b**, and can therefore affect the point in time at which the inverters **628a**, **628b** switch during transitions between signaling states **646a**, **646b**.

(53) The RC constant provided by the feedback resistor **634** and the input capacitor **632** effectively partially flattens the transitions in the Inv.sub.mid signal **642** between signaling states **646a**, **646b**. The slower transitions in the Inv.sub.mid signal **642** change the effective threshold of the inverters **628a**, **628b** and can amplify the effect of the raised voltage levels in the Inv.sub.mid signal **642**. In the example illustrated by the timing diagram **650**, the duty cycle distortion is almost completely corrected by the self-biasing inverter **630**. The high signaling state **648a** and the low signaling state **648b** have durations that are close to nominal values for a 50% duty cycle.

(54) The variable capacitor **606** in the programmable clock conditioning circuit **600** adds programmability to the duty cycle correction circuit **608** and can improve the effectiveness of the skew correction circuit **604** in addition to improving the effectiveness of the duty cycle correction circuit **608**. The variable capacitor **606** can be configured to increase transition times between signaling states **646a**, **646b** of a clock signal and can facilitate the operation of the skew correction circuit **604** and the duty cycle correction circuit **608** when correcting skew and duty cycle errors in the complementary output signals **620a**, **620b** output by the single-ended to differential signal conversion circuit **602**.

(55) In the illustrated example, the variable capacitor **606** includes two sets of capacitors **624a**, **624b** coupled through a switch matrix **626**. When activated, each switch in the switch matrix **626** is configured to couple the complementary signals **614a**, **614b** output by the single-ended to differential signal conversion circuit **602** through a series-connected combination of a capacitor in the first set of capacitors **624a** and a corresponding capacitor in the second set of capacitors **624b**. The switches in the switch matrix **626** are activated or deactivated by a switch control signal **622**. In various examples, the switch control signal **622** encodes a multibit value using binary-weighted or unary encoding. Unary encoding, which may be referred to as thermometer encoding, represents data in the quantity of bits set to '1' that precede a terminating '0', or the quantity of bits set to '0' that precede a terminating '1'.

(56) The skew and duty cycle corrections provided by the programmable clock conditioning circuit **600** can be increased by increasing the capacitance provided by the variable capacitor **606**.

Increasing the capacitance of the variable capacitor **606** tends to increase power consumption.

However, such increased power consumption is offset by the overall reduced power consumption and reduced die area obtained through the use of single stage skew and duty cycle correction, with respect to conventional multi-stage duty cycle correction and skew correction circuits. The programmable clock conditioning circuit **600** can be implemented and operated without calibration.

(57) FIG. 7 illustrates an example of a single-ended programmable clock conditioning circuit **700** that has been configured in accordance with certain aspects of this disclosure. The single-ended

programmable clock conditioning circuit **700** receives a single-ended input clock signal (the Clk_in signal **710**) and outputs a single-ended output clock signal **730**. The single-ended programmable clock conditioning circuit **700** includes a configurable inverter **702**, a variable capacitor **706** and a duty cycle correction circuit **708**.

(58) Certain output switching characteristics of the configurable inverter **702** may be configured, or controlled by adjusting the current supplied to the configurable inverter **702**. In the illustrated example, current supply to the configurable inverter **702** is controlled using one or more PMOS transistors **704a** to couple the configurable inverter **702** to the higher voltage power supply rail (V.sub.DD) and one or more NMOS transistors **704b** to couple the configurable inverter **702** to the lower voltage power supply rail (Ground). Current flow can be controlled by increasing the number of transistors **704a**, **704b** that actively couple the configurable inverter **702** to a power supply rail. In the illustrated example, the number of activated PMOS transistors **704a** and activated NMOS transistors **704b** is determined by an N-bit control code provided in a driver control signal **728**.

(59) Transitions in the Clk_in signal **710** may be slowed by an amount that is determined in part by the current levels provided to the configurable inverter **702**. The delay applied to the Clk_in signal **710** may be controlled by adjusting the configuration of the transistors **704a**, **704b**. Transitions in the Clk_in signal **710** may be further slowed based on the RC constant associated with the variable capacitor **706** and a duty cycle correction circuit **708**.

(60) In the illustrated example, the duty cycle correction circuit **708** is implemented using the self-biasing inverter **630** illustrated in FIG. 6. The output (i.e., the Clk_out signal **720**) of the duty cycle correction circuit **708** is provided to an inverter **712** that provides the single-ended output clock signal **730**. The duty cycle correction circuit **708** is programmable. The delayed clock signal **714** output by the configurable inverter **702** is coupled to the gates of a pair of transistors **708a**, **708b** through an input capacitor **718**. The DC component of the delayed clock signal **714** is blocked by the input capacitor **718** to provide a clock signal (the Clk.sub.mid signal **736**) that is centered on the mid-level DC voltage of the delayed clock signal **714**. A feedback resistor **734** is coupled between the Clk_out signal **720** and the gates of the pair of transistors **708a**, **708b**. The feedback resistor **734** is configured to automatically bias the pair of transistors **708a**, **708b**.

(61) The RC constant provided by the feedback resistor **734** and the input capacitor **718** is selected to provide a maximum transition time for signals in the Clk.sub.mid signal **736**. The mid-level voltage of the Clk.sub.mid signal **736** rises above the mid-level voltage of the delayed clock signal **714** due to the imbalance of charge accumulation during the different half cycles of the delayed clock signal **714**. The rise in the Clk.sub.mid signal **736** tends to drive the duty cycle of the Clk_out signal **720** towards 50%, reflecting a more balanced charge accumulation between signaling states. The raised voltage levels of the Clk.sub.mid signal **736** with respect to the delayed clock signal **714** changes the relationship of the Clk.sub.mid signal **736** with the switching thresholds associated with the inverter **712**, and can therefore affect the point in time at which the inverter **712** switches during transitions between signaling states.

(62) The RC constant provided by the feedback resistor **734** and the input capacitor **718** effectively flattens the transitions in the Clk.sub.mid signal **736** between signaling states. The slower transitions in the Clk.sub.mid signal **736** change the effective switching threshold of the transistors **708a**, **708b** and can amplify the effect of the raised voltage levels in the Clk.sub.mid signal **736**.

(63) The variable capacitor **706** in the single-ended programmable clock conditioning circuit **700** adds programmability to the duty cycle correction circuit **708** and can improve the effectiveness of the duty cycle correction circuit **708**. The variable capacitor **706** can be configured to reduce transition times between signaling states of a clock signal and can facilitate the operation of the duty cycle correction circuit **708**.

(64) In the illustrated example, the variable capacitor **706** includes a set of capacitors **724** coupled to a switch matrix **726**. When activated, each switch in the switch matrix **726** is configured to couple the delayed clock signal **714** to ground through a corresponding capacitor in the set of

capacitors **724**. The switches in the switch matrix **726** are activated or deactivated by a switch control signal **722**. In various examples, the switch control signal **722** encodes a multibit value using binary-weighted or unary encoding. In some instances, each switch in the switch matrix **726** can be configured to couple the delayed clock signal **714** to V.sub.DD through a corresponding capacitor in the set of capacitors **724** when activated.

(65) The duty cycle correction provided by the single-ended programmable clock conditioning circuit **700** can be increased by increasing the capacitance provided by the variable capacitor **706**. Increasing the capacitance of the variable capacitor **706** tends to increase power consumption. However, such increased power consumption is offset by the overall reduced power consumption and reduced die area obtained through the use of single stage duty cycle correction, and offer flexibility to save power and die area, with respect to conventional multi-stage duty cycle correction circuits. The single-ended programmable clock conditioning circuit **700** can be implemented and operated without calibration.

(66) FIG. **8** is a flowchart **8** of a method **800** for conditioning clock signals. In some examples, the method **800** may be implemented using the programmable clock conditioning circuit **600** illustrated in FIG. **6** or the single-ended programmable clock conditioning circuit **700** illustrated in FIG. **7**.

(67) At block **802** in the illustrated method, a first clock signal may be provided to an input of a duty cycle correction circuit, the duty cycle correction circuit having a first self-biasing inverter and a first input capacitor. At block **804** in the illustrated method, a capacitance value provided by a variable capacitor may be selected. The variable capacitor may be coupled to the first input capacitor. A capacitance provided by the variable capacitor may be programmable using a capacitance control signal. In one example, the variable capacitor may be coupled between the clock signal and a power supply rail. In another example, the variable capacitor may be coupled between complementary signals in a differential signal. The capacitance control signal may encode a multibit value used to select the capacitance provided by the variable capacitor. In one example, the capacitance control signal is a multi-bit unary or binary encoded signal.

(68) In certain implementations, a second clock signal is provided to the duty cycle correction circuit. The duty cycle correction circuit may have a second self-biasing inverter and a second input capacitor, and the variable capacitor may be coupled between the first clock signal and the second clock signal. A single-ended input clock signal may be provided to an input circuit configured to convert the single-ended input clock signal to a differential clock signal. The differential clock signal may include the first clock signal and the second clock signal. The first clock signal and the second clock signal may be complementary signals. A skew correction circuit may be provided between the first clock signal and the second clock signal and may include at least one pair of back-to-back inverters. The input of each back-to-back inverter in a pair of back-to-back inverters is coupled to the output of the other back-to-back inverter in the pair of back-to-back inverters. The capacitance value provided by the variable capacitor may be selected by activating one or more switches controlled by the capacitance control signal. Each of the one or more switches may be configured to couple the first clock signal and the second clock signal through a corresponding pair of series connected differential capacitors when activated. In one example, a first differential capacitor is coupled between the first clock signal and a first terminal of a first switch and a second differential capacitor is coupled between the second clock signal and a second terminal of the first switch.

(69) In some implementations, first self-biasing inverter has a feedback resistor that is coupled between the input of the first self-biasing inverter and an output of the first self-biasing inverter. In one example, the first self-biasing inverter is implemented using a push-pull amplifier in which the feedback resistor is coupled between the gates and the drains of the transistors in the push-pull amplifier.

(70) In certain implementations, a configurable inverter provides the first clock signal by delaying an input clock signal. In some of these implementations, the variable capacitor is coupled between

the first clock signal and ground. In these implementations, the capacitance value provided by the variable capacitor may be selected by activating one or more switches controlled by the capacitance control signal. Each of the one or more switches is configured to couple the first clock signal to ground through one of a plurality of capacitors provided in the variable capacitor when activated. (71) The method illustrated in FIG. 8 may be executed using a circuit provided in an IC device. In one example, the IC device comprises an SoC. In another example, the IC device is included in one of multiple semiconductor dice mounted on a substrate. The IC device may be implemented as a chiplet in some instances.

(72) The IC device may include a bus interface. The bus interface may include a clock conditioning circuit that includes a duty cycle correction circuit. The duty cycle correction circuit may include a first input capacitor and a first self-biasing inverter having an input coupled to the first input capacitor. The duty cycle correction circuit may include a variable capacitor coupled to the first input capacitor. The variable capacitor may be configured to receive a first clock signal. A capacitance of the variable capacitor may be programmable by a capacitance control signal. In various examples, the capacitance control signal is a multi-bit unary or binary encoded signal.

(73) In certain implementations, the duty cycle correction circuit includes a second input capacitor and a second self-biasing inverter. The second self-biasing inverter has an input coupled to the second clock signal through the second input capacitor. In these implementations, the variable capacitor is coupled between the first clock signal and the second clock signal. An input circuit and/or converter circuit may be configured to receive a single-ended input clock signal and to convert the single-ended input clock signal to a differential clock signal. The first clock signal and the second clock signal may be complementary signals provided in the differential clock signal. A skew correction circuit may be provided between the first clock signal and the second clock signal and may include at least one pair of back-to-back inverters. In some examples, the variable capacitor has a plurality of switches controlled by the capacitance control signal. Each activated switch may be configured to couple the first clock signal and the second clock signal through a corresponding pair of series connected differential capacitors.

(74) In some implementations, first self-biasing inverter has a feedback resistor that is coupled between the input of the first self-biasing inverter and an output of the first self-biasing inverter. In one example, the first self-biasing inverter is implemented using a push-pull amplifier in which the feedback resistor is coupled between the gates and the drains of the transistors in the push-pull amplifier.

(75) In certain implementations, the clock conditioning circuit includes a configurable inverter that is configured to provide the first clock signal by delaying an input clock signal. In these implementations, the variable capacitor may be coupled between the first clock signal and ground. In one example, the variable capacitor includes a plurality of switches controlled by the capacitance control signal. Each switch is configured to couple the first clock signal to ground through one of a plurality of capacitors provided in the variable capacitor when the switch is activated.

(76) Some implementation examples are described in the following numbered clauses: 1. A clock conditioning circuit, comprising: a duty cycle correction circuit, including a first input capacitor, and a first self-biasing inverter having an input coupled to the first input capacitor; and a variable capacitor coupled to the first input capacitor, the variable capacitor configured to receive a first clock signal, a capacitance of the variable capacitor being programmable by a capacitance control signal. 2. The clock conditioning circuit as described in clause 1, wherein the duty cycle correction circuit comprises: a second input capacitor; and a second self-biasing inverter having an input coupled to a second clock signal through the second input capacitor, wherein the variable capacitor is coupled between the first clock signal and the second clock signal. 3. The clock conditioning circuit as described in clause 2, further comprising: an input circuit configured to receive a single-ended input clock signal and to convert the single-ended input clock signal to a differential clock signal that comprises the first clock signal and the second clock signal; and a skew correction

circuit comprising at least one pair of back-to-back inverters coupled between the first clock signal and the second clock signal. 4. The clock conditioning circuit as described in clause 2, or clause 3 wherein the variable capacitor comprises: a plurality of switches controlled by the capacitance control signal, wherein each switch is configured to couple the first clock signal and the second clock signal through a corresponding pair of series connected differential capacitors when activated. 5. The clock conditioning circuit as described in any of clauses 1-4, wherein the capacitance control signal comprises a multi-bit unary or binary encoded signal. 6. The clock conditioning circuit as described in any of clauses 1-5, wherein the first self-biasing inverter comprises a feedback resistor coupled between the input of the first self-biasing inverter and an output of the first self-biasing inverter. 7. The clock conditioning circuit as described in clause 1, further comprising: a configurable inverter configured to provide the first clock signal by delaying an input clock signal, wherein the variable capacitor is coupled between the first clock signal and ground. 8. The clock conditioning circuit as described in clause 7, wherein the variable capacitor comprises: a plurality of switches controlled by the capacitance control signal, wherein each switch is configured to couple the first clock signal to ground through one of a plurality of capacitors provided in the variable capacitor when the each switch is activated. 9. A method for conditioning clock signals, comprising: providing a first clock signal to an input of a duty cycle correction circuit, the duty cycle correction circuit having a first self-biasing inverter and a first input capacitor; and selecting a capacitance value provided by a variable capacitor that is coupled to the first input capacitor, a capacitance provided by the variable capacitor being programmable by a capacitance control signal. 10. The method as described in clause 9, further comprising: providing a second clock signal to the duty cycle correction circuit wherein the duty cycle correction circuit comprises a second self-biasing inverter and a second input capacitor, wherein the variable capacitor is coupled between the first clock signal and the second clock signal. 11. The method as described in clause 10, further comprising: providing a single-ended input clock signal to an input circuit configured to convert the single-ended input clock signal to a differential clock signal that comprises the first clock signal and the second clock signal; and correcting skew between the first clock signal and the second clock signal using at least one pair of back-to-back inverters coupled between the first clock signal and the second clock signal. 12. The method as described in clause 10 or clause 11, wherein selecting the capacitance value provided by the variable capacitor comprises: activating one or more switches controlled by the capacitance control signal, wherein each of the one or more switches is configured to couple the first clock signal and the second clock signal through a corresponding pair of series connected differential capacitors when activated. 13. The method as described in any of clauses 9-12, wherein the capacitance control signal comprises a multi-bit unary or binary encoded signal. 14. The method as described in any of clauses 9-13, wherein the first self-biasing inverter comprises a feedback resistor coupled between the input of the first self-biasing inverter and an output of the first self-biasing inverter. 15. The method as described in clause 9, further comprising: causing a configurable inverter to provide the first clock signal by delaying an input clock signal, wherein the variable capacitor is coupled between the first clock signal and ground. 16. The method as described in clause 15, wherein selecting the capacitance value provided by the variable capacitor comprises: activating one or more switches controlled by the capacitance control signal, wherein each of the one or more switches is configured to couple the first clock signal to ground through one of a plurality of capacitors provided in the variable capacitor when activated. 17. An apparatus, comprising: means for correcting a duty cycle of a clock signal, including a first self-biasing inverter and a first input capacitor; means for adjusting transition times of edges in the clock signal, including a variable capacitor that is coupled to the first input capacitor; and means for programming a capacitance value provided by a variable capacitor, the means for programming the capacitance value being responsive to a capacitance control signal. 18. The apparatus as described in clause 17, wherein the means for correcting the duty cycle of the first clock signal includes: a second self-biasing inverter

and a second input capacitor, wherein the variable capacitor is coupled between the first input capacitor and the second input capacitor. 19. The apparatus as described in clause 18, wherein the means for correcting the duty cycle of the first clock signal includes: an input circuit configured to receive a single-ended input clock signal and to convert the single-ended input clock signal to a differential clock signal that comprises a first complementary signal and the second complementary signal; and a skew correction circuit comprising at least one pair of back-to-back inverters coupled between the first complementary signal and the second complementary signal. 20. The apparatus as described in clause 17, wherein the means for correcting the duty cycle of the first clock signal includes: a configurable inverter configured to couple the clock signal to the first input capacitor and further configured to delay the clock signal, and wherein the variable capacitor is coupled between the clock signal and ground.

(77) It is understood that the specific order or hierarchy of steps in the processes disclosed is an illustration of exemplary approaches. Based upon design preferences, it is understood that the specific order or hierarchy of steps in the processes may be rearranged. Further, some steps may be combined or omitted. The accompanying method claims present elements of the various steps in a sample order, and are not meant to be limited to the specific order or hierarchy presented.

(78) The previous description is provided to enable any person skilled in the art to practice the various aspects described herein. Various modifications to these aspects will be readily apparent to those skilled in the art, and the generic principles defined herein may be applied to other aspects. Thus, the claims are not intended to be limited to the aspects shown herein, but is to be accorded the full scope consistent with the language claims, wherein reference to an element in the singular is not intended to mean “one and only one” unless specifically so stated, but rather “one or more.” Unless specifically stated otherwise, the term “some” refers to one or more. All structural and functional equivalents to the elements of the various aspects described throughout this disclosure that are known or later come to be known to those of ordinary skill in the art are expressly incorporated herein by reference and are intended to be encompassed by the claims. Moreover, nothing disclosed herein is intended to be dedicated to the public regardless of whether such disclosure is explicitly recited in the claims. No claim element is to be construed as a means plus function unless the element is expressly recited using the phrase “means for.”

Claims

1. A clock conditioning circuit, comprising: a duty cycle correction circuit, including a first input capacitor, and a first self-biasing inverter having an input coupled to the first input capacitor; and a variable capacitor coupled to the first input capacitor, the variable capacitor configured to receive a first clock signal, a capacitance of the variable capacitor being programmable by a capacitance control signal.
2. The clock conditioning circuit of claim 1, wherein the duty cycle correction circuit comprises: a second input capacitor; and a second self-biasing inverter having an input coupled to a second clock signal through the second input capacitor, wherein the variable capacitor is coupled between the first clock signal and the second clock signal.
3. The clock conditioning circuit of claim 2, further comprising: an input circuit configured to receive a single-ended input clock signal and to convert the single-ended input clock signal to a differential clock signal that comprises the first clock signal and the second clock signal; and a skew correction circuit comprising at least one pair of back-to-back inverters coupled between the first clock signal and the second clock signal.
4. The clock conditioning circuit of claim 2, wherein the variable capacitor comprises: a plurality of switches controlled by the capacitance control signal, wherein each switch is configured to couple the first clock signal and the second clock signal through a corresponding pair of series connected differential capacitors when activated.

5. The clock conditioning circuit of claim 1, wherein the capacitance control signal comprises a multi-bit unary or binary encoded signal.
6. The clock conditioning circuit of claim 1, wherein the first self-biasing inverter comprises a feedback resistor coupled between the input of the first self-biasing inverter and an output of the first self-biasing inverter.
7. The clock conditioning circuit of claim 1, further comprising: a configurable inverter configured to provide the first clock signal by delaying an input clock signal, wherein the variable capacitor is coupled between the first clock signal and ground.
8. The clock conditioning circuit of claim 7, wherein the variable capacitor comprises: a plurality of switches controlled by the capacitance control signal, wherein each switch is configured to couple the first clock signal to ground through one of a plurality of capacitors provided in the variable capacitor when the each switch is activated.
9. A method for conditioning clock signals, comprising: providing a first clock signal to an input of a duty cycle correction circuit, the duty cycle correction circuit having a first self-biasing inverter and a first input capacitor; and selecting a capacitance value provided by a variable capacitor that is coupled to the first input capacitor, a capacitance provided by the variable capacitor being programmable by a capacitance control signal.
10. The method of claim 9, further comprising: providing a second clock signal to the duty cycle correction circuit wherein the duty cycle correction circuit comprises a second self-biasing inverter and a second input capacitor, wherein the variable capacitor is coupled between the first clock signal and the second clock signal.
11. The method of claim 10, further comprising: providing a single-ended input clock signal to an input circuit configured to convert the single-ended input clock signal to a differential clock signal that comprises the first clock signal and the second clock signal; and correcting skew between the first clock signal and the second clock signal using at least one pair of back-to-back inverters coupled between the first clock signal and the second clock signal.
12. The method of claim 10, wherein selecting the capacitance value provided by the variable capacitor comprises: activating one or more switches controlled by the capacitance control signal, wherein each of the one or more switches is configured to couple the first clock signal and the second clock signal through a corresponding pair of series connected differential capacitors when activated.
13. The method of claim 9, wherein the capacitance control signal comprises a multi-bit unary or binary encoded signal.
14. The method of claim 9, wherein the first self-biasing inverter comprises a feedback resistor coupled between the input of the first self-biasing inverter and an output of the first self-biasing inverter.
15. The method of claim 9, further comprising: causing a configurable inverter to provide the first clock signal by delaying an input clock signal, wherein the variable capacitor is coupled between the first clock signal and ground.
16. The method of claim 15, wherein selecting the capacitance value provided by the variable capacitor comprises: activating one or more switches controlled by the capacitance control signal, wherein each of the one or more switches is configured to couple the first clock signal to ground through one of a plurality of capacitors provided in the variable capacitor when activated.
17. An apparatus, comprising: means for correcting a duty cycle of a clock signal, including a first self-biasing inverter and a first input capacitor; means for adjusting transition times of edges in the clock signal, including a variable capacitor that is coupled to the first input capacitor; and means for programming a capacitance value provided by a variable capacitor, the means for programming the capacitance value being responsive to a capacitance control signal, wherein the means for correcting the duty cycle of the clock signal includes: a second self-biasing inverter and a second input capacitor, wherein the variable capacitor is coupled between the first input capacitor and the

second input capacitor.

18. The apparatus of claim 17, wherein the means for correcting the duty cycle of the clock signal includes: an input circuit configured to receive a single-ended input clock signal and to convert the single-ended input clock signal to a differential clock signal that comprises a first complementary signal and a second complementary signal; and a skew correction circuit comprising at least one pair of back-to-back inverters coupled between the first complementary signal and the second complementary signal.

19. An apparatus, comprising: means for correcting a duty cycle of a clock signal, including a first self-biasing inverter and a first input capacitor; means for adjusting transition times of edges in the clock signal, including a variable capacitor that is coupled to the first input capacitor; and means for programming a capacitance value provided by a variable capacitor, the means for programming the capacitance value being responsive to a capacitance control signal wherein the means for correcting the duty cycle of the clock signal includes: a configurable inverter configured to couple the clock signal to the first input capacitor and further configured to delay the clock signal, and wherein the variable capacitor is coupled between the clock signal and ground.
